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A Appendix

Additional Figures

In this main paper, we report results based on models that are trained to optimize the $N = 1$ single token-ahead prediction, and we test those models for predictive accuracy for other N.

The same methods can also be used to optimize subsequent tokens, and the results of those methods are shown here. We find that optimizing for $N = 1$ works best and generalizes surprisingly well to other N, but that that optimizing for other N does not perform well for $N = 1$.

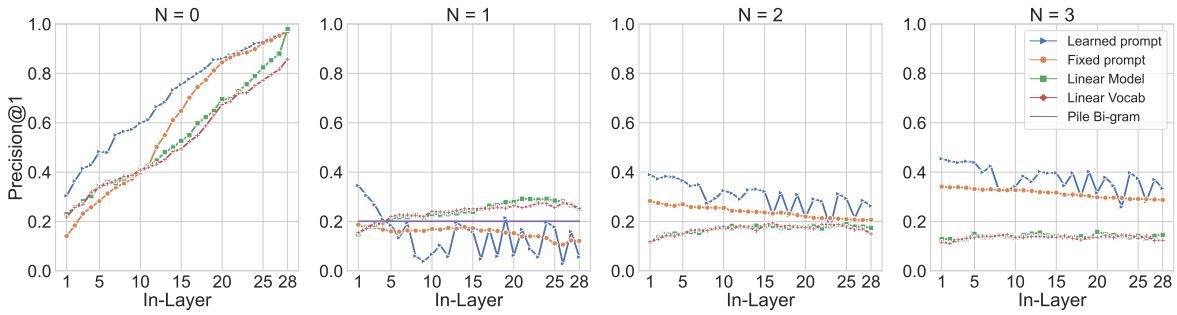

Figure 7: The Precision@1 (Accuracy) of all the methods trained with predicting the currently decoded token (teacher-forcing)

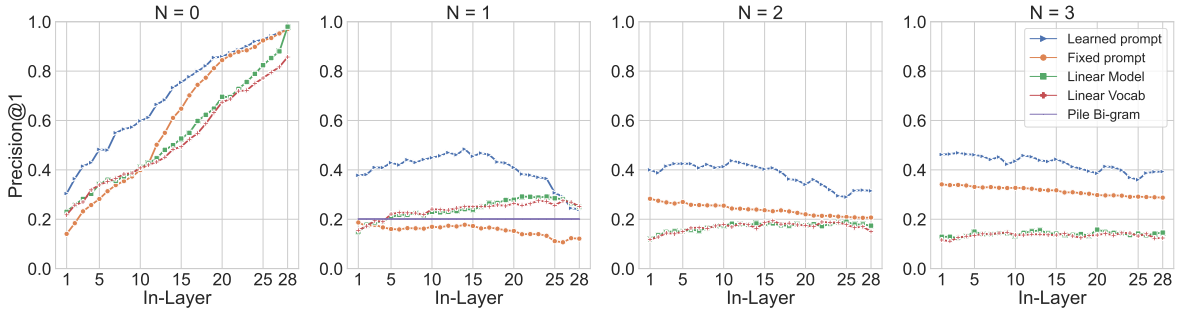

Figure 8: The Precision@1 (Accuracy) of all the methods trained with predicting the 1st next token

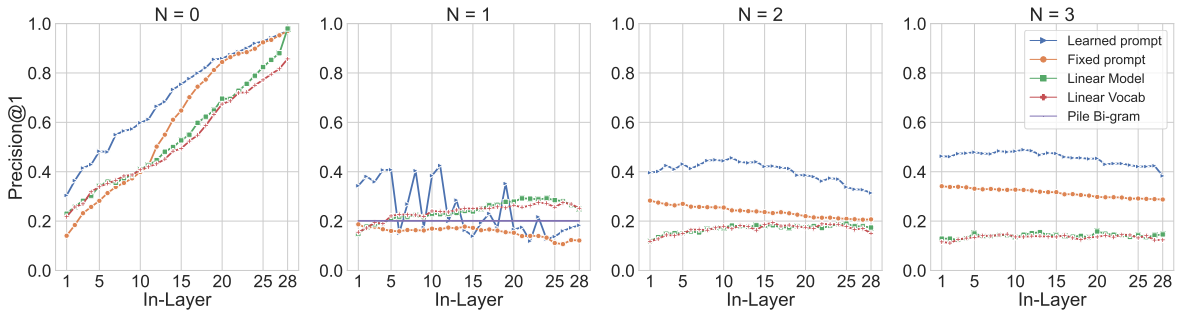

Figure 9: The Precision@1 (Accuracy) of all the methods trained with predicting the 2nd next token

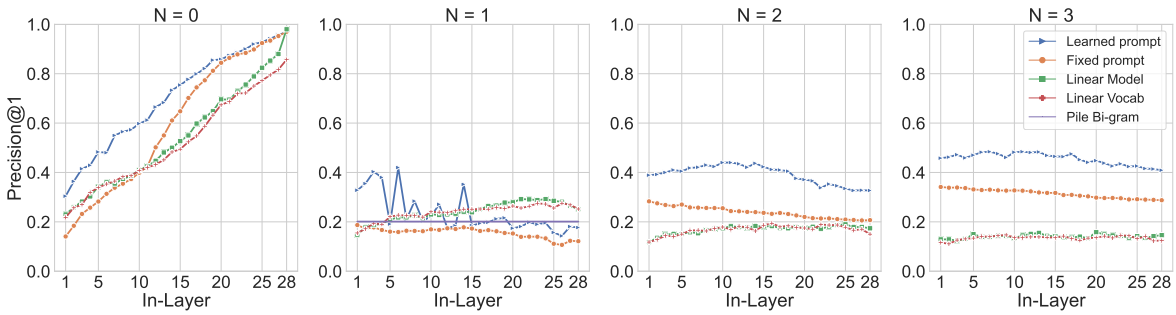

Figure 10: The Precision@1 (Accuracy) of all the methods trained with predicting the 3rd next token

Limitations

In our exploration with extracting far future tokens from single hidden states, we have mostly trained and tested on English data whose size, 100,000, is still relatively small compared to the data size that GPT-J-6B was actually trained in. Furthermore, the experiments were only conducted in GPT-J-6B. While the presence of subsequent token information in a single hidden state is evident in this model, it would be more comprehensive to run these experiments in other LLMs. Since there are no specific prior works that focused on decoding far future tokens from a single hidden state, we did not have any prior baselines we would refer to. While we did create a bigram baseline in the case of predicting 2 tokens in the future ($N = 1$) and also create linear models as a first decoding method, there could be baselines with other architectures like Recurrent Neural Networks (Jordan, 1997; Elman, 1990) and Non-Autoregressive generation (Su et al., 2021; Xiao et al., 2023). Lastly, our experiments were up to 4 tokens in the future, i.e., $N = 0, 1, 2, 3$. It would be intriguing to scale and test up to how many tokens in the future does a single state actually encode and predict.